

Course Review: Group Theory and Galois Theory

MAT205 Abstract Algebra II

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Part I: Group Theory

Sylow's Theorems

Definition. A Sylow p -subgroup of G is a maximal p -subgroup. For $|G| = p^a m$ with $\gcd(p, m) = 1$: P is Sylow $\iff |P| = p^a$. Write $\text{Syl}_p(G)$ for the set of Sylow p -subgroups, $n_p = |\text{Syl}_p(G)|$.

Cauchy's Theorem (1845). If $p \mid |G|$, then G has an element of order p .

Sylow I. If $p^i \mid |G|$ ($1 \leq i \leq a$), G has a subgroup of order p^i ; in particular Sylow p -subgroups exist.

Sylow II. All Sylow p -subgroups of G are conjugate.

Sylow III. $n_p \mid m$ and $n_p \equiv 1 \pmod{p}$.

Corollary. $P \trianglelefteq G \iff n_p = 1$. So to show G is *not* simple, it suffices to force $n_p = 1$ for some prime p .

Worked Example: Sylow Counting Forces a Normal Subgroup

Claim. No group of order 80 is simple.

Proof. $|G| = 80 = 2^4 \cdot 5$. By Sylow III, $n_5 \mid 16$ and $n_5 \equiv 1 \pmod{5}$; among the divisors 1, 2, 4, 8, 16 of 16, only 1 and 16 are $\equiv 1 \pmod{5}$. So $n_5 \in \{1, 16\}$.

Case $n_5 = 1$. The unique Sylow 5-subgroup is normal. Done.

Case $n_5 = 16$. Each Sylow 5-subgroup has prime order 5, so distinct ones intersect trivially: $16 \cdot 4 = 64$ elements of order 5. That leaves $80 - 64 = 16$ elements, exactly the size of *one* Sylow 2-subgroup, which must be the unique (hence normal) one.

Either way G has a nontrivial proper normal subgroup. □

Worked Example: Two Sylow Counts Jointly Overflow

Claim. No group of order 105 is simple.

Proof. $|G| = 105 = 3 \cdot 5 \cdot 7$. By Sylow III: $n_7 \mid 15$, $n_7 \equiv 1 \pmod{7} \Rightarrow n_7 \in \{1, 15\}$; and $n_5 \mid 21$, $n_5 \equiv 1 \pmod{5} \Rightarrow n_5 \in \{1, 21\}$.

Suppose $n_7 = 15$ *and* $n_5 = 21$. Sylow 7- and 5-subgroups have prime order, so they pairwise intersect trivially, contributing $15 \cdot 6 = 90$ elements of order 7 and $21 \cdot 4 = 84$ elements of order 5 — disjoint sets, needing $90 + 84 = 174 > 105$ elements. Impossible.

So $n_7 = 1$ or $n_5 = 1$: a normal Sylow subgroup either way.



p -Groups: Nontrivial Center and Order p^2

Theorem. If $|G| = p^k > 1$ (p prime), then $Z(G) \neq \{e\}$.

Corollary. Every group of order p^2 is abelian.

Proof. $Z(G) \neq \{e\}$ and $Z(G) \leq G$, so by Lagrange $|Z(G)| \in \{p, p^2\}$.

- If $|Z(G)| = p$: then $G/Z(G)$ has order p , hence is cyclic. A group with cyclic $G/Z(G)$ is abelian, forcing $Z(G) = G$ — contradicting $|Z(G)| = p < p^2 = |G|$.
- So $|Z(G)| = p^2$, i.e. $Z(G) = G$, i.e. G is abelian. □

Fundamental Theorem of Finite Abelian Groups

Theorem. Every finite abelian group G is isomorphic to

$$\mathbb{Z}/d_1 \oplus \mathbb{Z}/d_2 \oplus \cdots \oplus \mathbb{Z}/d_k, \quad d_1 \mid d_2 \mid \cdots \mid d_k$$

(invariant factors), equivalently to a direct sum of cyclic groups of prime-power order (elementary divisors). This decomposition is unique up to reordering.

Equivalently: G decomposes canonically as $\bigoplus_p T_{p^\infty}(G)$, the direct sum of its Sylow (primary) subgroups, and each primary piece is a direct sum of cyclic p -power groups.

Worked Example: Classifying Abelian Groups of a Given Order

Problem. List all abelian groups of order 200, up to isomorphism.

Solution. $200 = 2^3 \cdot 5^2$.

- 2-part: one of $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, $\mathbb{Z}_2^3$ (the 3 partitions of 3).
- 5-part: one of $\mathbb{Z}_{25}$, $\mathbb{Z}_5 \times \mathbb{Z}_5$ (the 2 partitions of 2).

The six groups are

$$\begin{aligned} &\mathbb{Z}_8 \times \mathbb{Z}_{25}, \quad \mathbb{Z}_8 \times \mathbb{Z}_5^2, \quad \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}, \\ &\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_5^2, \quad \mathbb{Z}_2^3 \times \mathbb{Z}_{25}, \quad \mathbb{Z}_2^3 \times \mathbb{Z}_5^2. \end{aligned}$$

Composition Series and the Jordan–Hölder Theorem

Definition. A nontrivial group S is *simple* if its only normal subgroups are 1 and S .

Definition. A *composition series* is a subnormal series

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_n = 1$$

with every factor G_i/G_{i+1} simple; n is the *composition length*. Every finite group has one.

Jordan–Hölder Theorem. Any two composition series of G have the same length, and after reordering, isomorphic factors: the multiset $\{G_i/G_{i+1}\}$ of composition factors is an invariant of G .

Worked Example: Composition Factors of an Abelian Group

Problem. Determine the composition factors of $A = \mathbb{Z}_{20} \times \mathbb{Z}_{45}$.

Solution. A is finite abelian, so its composition factors are cyclic of prime order, one for each prime factor of $|A|$ (with multiplicity). Since

$$|A| = 20 \cdot 45 = 900 = 2^2 \cdot 3^2 \cdot 5^2,$$

the composition factors are $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_3, \mathbb{Z}_5, \mathbb{Z}_5$.

Solvable Groups

Definition. G is *solvable* if it has a subnormal series $G = G_0 \geq G_1 \geq \cdots \geq G_n = 1$ with each G_i/G_{i+1} abelian.

Theorem (finite groups).

G solvable $\iff$ every composition factor of G is cyclic of prime order

$\iff$ no composition factor of G is non-abelian simple.

A_n Is Simple for $n \geq 5$

Theorem. A_n is simple and non-abelian for every $n \geq 5$.

This fails below $n = 5$: $A_3 \cong \mathbb{Z}_3$ is abelian simple, and A_4 has a normal Klein-four subgroup $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$.

Worked Example: Proving a Non-Abelian Group Is Solvable

Problem. Show A_4 is solvable, and find its composition factors.

Solution.

$$A_4 \triangleright V_4 \triangleright \langle (12)(34) \rangle \triangleright 1$$

is a composition series ($V_4 \trianglelefteq A_4$; $\langle (12)(34) \rangle \trianglelefteq V_4$ since V_4 is abelian). The successive quotients are

$$A_4/V_4 \cong \mathbb{Z}_3, \quad V_4/\langle (12)(34) \rangle \cong \mathbb{Z}_2, \quad \langle (12)(34) \rangle \cong \mathbb{Z}_2.$$

So the composition factors are $\mathbb{Z}_3, \mathbb{Z}_2, \mathbb{Z}_2$ — all cyclic of prime order, so by the composition-factor criterion, A_4 is solvable.

Worked Example: A Symmetric Group That Is Not Solvable

Claim. S_6 is not solvable.

Proof. $S_6 \triangleright A_6 \triangleright 1$ is a composition series: A_6 is simple (previous theorem, $6 \geq 5$) and non-abelian, and $S_6/A_6 \cong \mathbb{Z}_2$.

Since A_6 is a non-abelian simple composition factor, S_6 is not solvable. □

Characteristic Subgroups

Definition. $H \leq G$ is *characteristic* in G ($H \text{ char } G$) if $\alpha(H) = H$ for every $\alpha \in \text{Aut}(G)$ (the automorphism group of G).

Theorem. If $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.

Proof. Let $g \in G$. Since $H \trianglelefteq G$, conjugation $c_g(x) = gxg^{-1}$ maps H onto H , so $c_g|_H \in \text{Aut}(H)$. Since $K \text{ char } H$, every automorphism of H fixes K setwise, so $c_g(K) = K$, i.e. $gKg^{-1} = K$. As $g \in G$ was arbitrary, $K \trianglelefteq G$. □

The Normalizer–Centralizer Theorem

Definitions. $N_G(H) = \{g \in G : gHg^{-1} = H\};$
 $Z_G(H) = \{g \in G : ghg^{-1} = h \ \forall h \in H\}.$

Theorem. $Z_G(H) \trianglelefteq N_G(H).$

Proof. Conjugation gives a homomorphism $\varphi : N_G(H) \rightarrow \text{Aut}(H)$, $\varphi(g)(h) = ghg^{-1}$ (well defined because $g \in N_G(H)$ means $gHg^{-1} = H$). Its kernel is exactly $Z_G(H)$. A kernel is normal in the domain, so $Z_G(H) \trianglelefteq N_G(H)$. $\square$

Consequence. $N_G(H)/Z_G(H) \hookrightarrow \text{Aut}(H).$

Part II: Fields and Galois Theory

Tower Law

Theorem. For a tower of fields $F \subseteq K \subseteq E$,

$$[E : F] = [E : K] \cdot [K : F].$$

Minimal Polynomial and Conjugates

Definition. For α algebraic over F , the *minimal polynomial* $m_\alpha \in F[x]$ is the monic generator of $\{f \in F[x] : f(\alpha) = 0\}$; it is irreducible, and $[F(\alpha) : F] = \deg m_\alpha$.

Definition. α, β are *conjugate over F* if $m_\alpha = m_\beta$.

Theorem (Conjugation Isomorphism, Fraleigh 48.3). If α, β are conjugate over F , there is a unique F -isomorphism $F(\alpha) \rightarrow F(\beta)$ sending $\alpha \mapsto \beta$.

Consequence. The conjugates of α over F are exactly the roots of m_α in $\overline{F}$.

Worked Example: Listing Conjugates

Problem. List the conjugates of $\sqrt[5]{11}$ over $\mathbb{Q}$.

Solution. $x^5 - 11$ is irreducible over $\mathbb{Q}$ (Eisenstein, $p = 11$), so it is the minimal polynomial of $\sqrt[5]{11}$. Its five roots in $\mathbb{C}$ are

$$\zeta_5^k \sqrt[5]{11}, \quad k = 0, 1, 2, 3, 4, \quad \zeta_5 = e^{2\pi i/5}.$$

These are exactly the conjugates of $\sqrt[5]{11}$ over $\mathbb{Q}$.

Worked Example: A Degree Computation

Problem. Compute $[\mathbb{Q}(\sqrt{6}, \sqrt[3]{10}) : \mathbb{Q}]$, with justification.

Solution. $[\mathbb{Q}(\sqrt{6}) : \mathbb{Q}] = 2$ and $[\mathbb{Q}(\sqrt[3]{10}) : \mathbb{Q}] = 3$ (irreducible by Eisenstein at $p = 2$ or $p = 5$). Writing $d = [\mathbb{Q}(\sqrt{6}, \sqrt[3]{10}) : \mathbb{Q}]$, the Tower Law applied through each subfield gives $2 \mid d$ and $3 \mid d$, so $6 \mid d$; the Tower Law also gives $d \leq 2 \cdot 3 = 6$. Hence $d = 6$.

Splitting Fields and Normal Extensions

Definition. $f \in F[x]$ *splits* in E if it factors into linear factors over E . E/F is a *splitting field* for f if f splits in E and E is generated over F by the roots of f .

Definition. E/F (inside a fixed $\overline{F}$) is *normal* if every F -embedding $\sigma : E \rightarrow \overline{F}$ satisfies $\sigma(E) = E$.

Theorem (equivalences, E/F finite). The following are equivalent: (1) every irreducible $p \in F[x]$ with a root in E splits in E ; (2) E is a splitting field over F of some $f \in F[x]$; (3) every F -embedding $E \rightarrow \overline{F}$ has image E .

Galois Extensions and the Galois Group

Definition/Recall. A finite extension E/F is *Galois* if it is normal and *separable* (minimal polynomials over F have no repeated roots — automatic in characteristic 0, and automatic for finite fields), equivalently

$$|\operatorname{Aut}(E/F)| = [E : F].$$

Its *Galois group* is

$$G := \operatorname{Gal}(E/F) := \operatorname{Aut}(E/F), \quad |G| = [E : F].$$

Worked Example: A Non-Normal Extension

Claim. $\mathbb{Q}(\sqrt[3]{3})$ is not a normal extension of $\mathbb{Q}$.

Proof. $x^3 - 3$ is irreducible over $\mathbb{Q}$ (Eisenstein, $p = 3$) with root $\sqrt[3]{3} \in \mathbb{Q}(\sqrt[3]{3})$, but its other two roots $\omega\sqrt[3]{3}$, $\omega^2\sqrt[3]{3}$ ($\omega = e^{2\pi i/3}$) are non-real, hence not in $\mathbb{Q}(\sqrt[3]{3}) \subset \mathbb{R}$.

An irreducible polynomial with one root in the field but not all its roots witnesses failure of normality. □

Finite Fields: Existence, Uniqueness, Multiplicative Group

Theorem. For each prime power $q = p^n$, there is a subfield $\mathbb{F}_q \subseteq \overline{\mathbb{F}_p}$ with exactly q elements, and any finite field of order q is isomorphic to it.

Theorem (Fraleigh 33.4). Every finite subgroup of $F^\times$ (any field F) is cyclic. In particular $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

The Frobenius Automorphism

Definition/Theorem. The Frobenius map $\sigma(a) = a^p$ is a field homomorphism $\mathbb{F}_q \rightarrow \mathbb{F}_q$. Since $\mathbb{F}_q$ is finite, σ is an automorphism.

Theorem. $\text{Aut}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \langle \sigma \rangle$ is cyclic of order n .

Subfields of $\mathbb{F}_{p^n}$; No Finite Field Is Algebraically Closed

Theorem (subfield criterion). $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n} \iff d \mid n$.

Theorem.

$$x^{p^n} - x = \prod_{a \in \mathbb{F}_{p^n}} (x - a).$$

Consequence. For $K = \mathbb{F}_q$, the polynomial $x^q - x + 1$ takes the value 1 at every $a \in K$, hence has no root in K : no finite field is algebraically closed.

Worked Example: Smallest Field Where $x^n - 1$ Splits

Problem. Find the smallest n such that $x^{15} - 1 \in \mathbb{F}_2[x]$ splits completely over $\mathbb{F}_{2^n}$.

Solution. $x^{15} - 1$ splits over $\mathbb{F}_{2^n}$ iff $\mathbb{F}_{2^n}^\times$ (cyclic of order $2^n - 1$) contains all 15th roots of unity, iff $15 \mid 2^n - 1$. Computing powers of 2 mod 15:

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8, \quad 2^4 = 16 \equiv 1 \pmod{15}.$$

The order of 2 mod 15 is 4, so the smallest such n is $n = 4$.

Cyclotomic Fields

Theorem. The minimal polynomial of $\zeta_n = e^{2\pi i/n}$ over $\mathbb{Q}$ is the cyclotomic polynomial Φ_n , irreducible of degree $\varphi(n)$. Hence

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n), \quad \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

via $\sigma_a : \zeta_n \mapsto \zeta_n^a$ for $a \in (\mathbb{Z}/n\mathbb{Z})^\times$.

Worked Example: Cyclotomic Galois Groups, Cyclic and Not

(a) Prime modulus. Is $\text{Gal}(\mathbb{Q}(\zeta_{11})/\mathbb{Q})$ cyclic?

Yes: $(\mathbb{Z}/11\mathbb{Z})^\times = \mathbb{F}_{11}^\times$ is the multiplicative group of the field $\mathbb{F}_{11}$, hence cyclic, of order 10.

(b) Composite modulus. Is $\text{Gal}(\mathbb{Q}(\zeta_{12})/\mathbb{Q})$ cyclic?

No: $\text{Gal}(\mathbb{Q}(\zeta_{12})/\mathbb{Q}) \cong (\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$, and every non-identity element squares to 1 mod 12:

$$5^2 = 25 \equiv 1, \quad 7^2 = 49 \equiv 1, \quad 11^2 = 121 \equiv 1 \pmod{12}.$$

So this group is $\mathbb{Z}_2 \times \mathbb{Z}_2$ — **not** cyclic.

Worked Example: Galois Group of a Cubic Splitting Field

Problem. Let E be the splitting field of $x^3 - 3$ over $\mathbb{Q}$. Find $\text{Gal}(E/\mathbb{Q})$ with explicit generators.

Solution. $E = \mathbb{Q}(\sqrt[3]{3}, \omega)$, $\omega = e^{2\pi i/3}$. By the Tower Law,

$$[E : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{3}) : \mathbb{Q}] \cdot [\mathbb{Q}(\sqrt[3]{3}, \omega) : \mathbb{Q}(\sqrt[3]{3})] = 3 \cdot 2 = 6.$$

E is a splitting field (normal) and separable (characteristic 0), so $E/\mathbb{Q}$ is Galois with $|\text{Gal}(E/\mathbb{Q})| = 6$. Let

$$\sigma : \sqrt[3]{3} \mapsto \omega \sqrt[3]{3}, \omega \mapsto \omega \text{ (order 3),} \quad \tau : \sqrt[3]{3} \mapsto \sqrt[3]{3}, \omega \mapsto \omega^2 \text{ (order 2).}$$

Then $\tau\sigma\tau^{-1} = \sigma^{-1}$, so $\langle \sigma, \tau \rangle \cong S_3$; since $|\langle \sigma, \tau \rangle| = 6 = |\text{Gal}(E/\mathbb{Q})|$, $\text{Gal}(E/\mathbb{Q}) = \langle \sigma, \tau \rangle \cong S_3$.

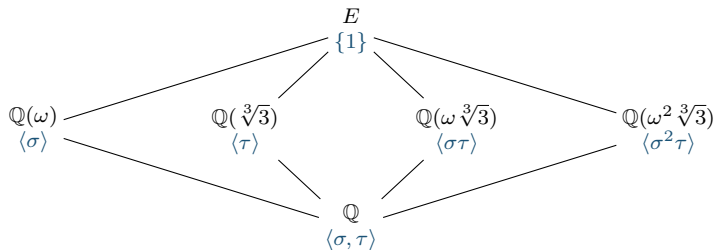
Fundamental Theorem of Galois Theory

Theorem. Let E/F be finite Galois, $G = \text{Gal}(E/F)$. The map $K \mapsto \text{Gal}(E/K)$ is a bijection from intermediate fields $F \subseteq K \subseteq E$ onto subgroups of G :

1. $K = E^{\text{Gal}(E/K)}$, and $\text{Gal}(E/E^H) = H$ for every $H \leq G$;
2. $[E : K] = |\text{Gal}(E/K)|$ and $[K : F] = [G : \text{Gal}(E/K)]$;
3. K/F is normal $\iff \text{Gal}(E/K) \trianglelefteq G$, and then $\text{Gal}(K/F) \cong G / \text{Gal}(E/K)$;
4. the correspondence reverses inclusions: bigger field $\leftrightarrow$ smaller subgroup.

Worked Example: Explicit Galois Correspondence

For $E = \mathbb{Q}(\sqrt[3]{3}, \omega)$ and σ, τ as before, the subgroup lattice of $\text{Gal}(E/\mathbb{Q}) = \langle \sigma, \tau \rangle \cong S_3$ and its fixed fields:



Only $\langle \sigma \rangle$ (order 3, index 2) is normal in $\langle \sigma, \tau \rangle$ — matching $\mathbb{Q}(\omega)/\mathbb{Q}$ being the unique normal proper subextension. The three order-2 subgroups are conjugate, not normal, matching the three non-normal cubic subfields.

Counting Irreducible Polynomials via Frobenius Orbits

Claim. The number of monic irreducible quadratics over $\mathbb{F}_q$ is $\frac{q^2 - q}{2}$.

Proof. $[\mathbb{F}_{q^2} : \mathbb{F}_q] = 2$ is prime, so any $\alpha \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ generates $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$: it has degree exactly 2, with minimal polynomial a monic irreducible quadratic over $\mathbb{F}_q$.

Let $\text{Fr}(x) = x^q$ generate $\text{Gal}(\mathbb{F}_{q^2}/\mathbb{F}_q) \cong \mathbb{Z}_2$. For such α , the orbit $\{\alpha, \alpha^q\}$ has size 1 or 2; size 1 forces $\alpha \in \mathbb{F}_q$, excluded. So every orbit in $\mathbb{F}_{q^2} \setminus \mathbb{F}_q$ has size 2, and each orbit is exactly the root set of one monic irreducible quadratic (and conversely).

$\mathbb{F}_{q^2} \setminus \mathbb{F}_q$ has $q^2 - q$ elements, partitioned into orbits of size 2: there are $\frac{q^2 - q}{2}$ such orbits, hence that many monic irreducible quadratics. □